

AI Model Comparison Report

Task 5 — Week 2, Resume AI Project

Overview

This report compares four language models that are commonly used for text understanding and similarity tasks: BERT, Sentence-BERT (SBERT), MiniLM, and DistilBERT. Each model is evaluated on its purpose, advantages, limitations, speed, accuracy, and suitability for this project.

Comparison Table

Model	Purpose	Advantages	Limitations	Speed	Recommended Use
BERT	General language understanding; produces word-level and sentence-level representations	Very strong at understanding context; widely used as a base for other models	Not designed to produce a single sentence embedding directly; comparing two sentences requires extra processing and is slow	Slow	Not ideal for this project's real-time similarity comparisons
SBERT	A modified version of BERT built specifically to produce meaningful sentence embeddings	Fast and accurate sentence comparison; embeddings can be directly compared with cosine similarity	Larger model size means more memory and processing power than smaller alternatives	Fast	Strong choice for comparing resumes and job descriptions
MiniLM	A smaller, distilled version of larger transformer models, built for efficiency	Very fast, low memory use, still fairly accurate for sentence similarity tasks	Slightly lower accuracy than larger models on complex or nuanced text	Very Fast	Good for quick similarity checks or when running on limited hardware
DistilBERT	A smaller, faster version of BERT, trained to keep most of BERT's understanding ability	Faster and lighter than BERT while keeping a large share of its accuracy	Still requires extra work to produce sentence-level embeddings, similar to BERT	Moderate	Reasonable general-purpose option, but not as suited to this project as SBERT

1. BERT

BERT (Bidirectional Encoder Representations from Transformers) was one of the first models to deeply understand context by reading text in both directions at once. It set the foundation for most modern language models, including the other three compared in this report.

- Purpose: General-purpose language understanding.

- Advantages: Strong contextual understanding of language.
- Limitations: Not built to directly output a single embedding representing a full sentence; comparing two sentences with BERT requires passing them through the model together, which is slow and not practical for comparing many resumes against many job descriptions.
- Speed: Slow, especially for large-scale comparisons.
- Accuracy: High, but at a high computational cost.
- Recommended use in this project: Not recommended directly, but its architecture is the foundation for SBERT and DistilBERT, which are better suited here.

2. Sentence-BERT (SBERT)

SBERT modifies BERT specifically to produce one meaningful embedding per sentence, allowing fast and direct comparison between two pieces of text using cosine similarity.

- Purpose: Efficient sentence and paragraph-level embeddings for comparison tasks.
- Advantages: Embeddings can be generated once and compared instantly; well-suited for search and matching tasks.
- Limitations: Larger than lightweight models like MiniLM, so it uses more memory and processing time per document.
- Speed: Fast, and far more efficient than BERT for comparison tasks.
- Accuracy: High accuracy on sentence similarity tasks.
- Recommended use in this project: This is the best fit for the resume-to-job-description similarity engine (Task 4), since the project needs accurate, direct comparison between two blocks of text.

3. MiniLM

MiniLM is a compact model designed to keep most of the performance of larger transformer models while being significantly smaller and faster.

- Purpose: Lightweight sentence embeddings for fast, resource-efficient comparison.
- Advantages: Very fast, low memory usage, works well even on limited hardware.
- Limitations: Slightly less accurate than larger models on complex or highly nuanced text.
- Speed: Very fast, among the fastest of the four.
- Accuracy: Good, though marginally lower than SBERT's larger variants.
- Recommended use in this project: A strong option if speed and low resource use are a priority, such as for quick preliminary scoring before a more detailed comparison.

4. DistilBERT

DistilBERT is a compressed version of BERT, trained to reproduce most of BERT's language understanding using a smaller model and fewer computations.

- Purpose: A faster, lighter alternative to BERT for general language tasks.
- Advantages: Meaningfully faster than BERT while retaining a large share of its accuracy.
- Limitations: Like BERT, it is not built to output a ready-to-compare sentence embedding on its own.
- Speed: Moderate — faster than BERT, slower than SBERT and MiniLM for this project's use case.
- Accuracy: Good general accuracy, close to BERT's.
- Recommended use in this project: A reasonable general-purpose option, but SBERT remains the better choice specifically for similarity comparison.

Conclusion

For this project's goal of comparing resumes against job descriptions using semantic similarity, SBERT is the recommended model. It is purpose-built for producing comparable sentence embeddings, offers a strong balance of speed and accuracy, and integrates directly with the sentence-transformers library already installed in the project environment. MiniLM is a good lightweight alternative if faster processing is needed at a small cost to accuracy.